

# Number theory – 5

## Equations in integers

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**Theoretical minimum.** Four methods exhaust nearly everything: reduce modulo a well-chosen number, factor over a larger ring, descend to a smaller solution, or bound the size of a solution.

**Theorem 1 (linear equations).**  $ax + by = c$  has an integer solution if and only if  $\gcd(a, b) \mid c$ , and then the solutions form an arithmetic progression  $x = x_0 + \frac{b}{\gcd}t$ ,  $y = y_0 - \frac{a}{\gcd}t$ . For several variables and congruence conditions the Chinese remainder theorem of Number theory – 1 reduces everything to prime powers.

**Theorem 2 (Pell's equation).** For a non-square positive integer  $d$  the equation  $x^2 - dy^2 = 1$  has infinitely many solutions in positive integers; if  $(x_1, y_1)$  is the least one, then all are given by  $x_n + y_n\sqrt{d} = (x_1 + y_1\sqrt{d})^n$ . *Proof:* Dirichlet's approximation theorem (Methods of proof – 1) produces infinitely many pairs with  $|x - y\sqrt{d}| < 1/y$ , hence  $|x^2 - dy^2| < 2\sqrt{d} + 1$ ; by pigeonhole some value  $k$  is taken infinitely often with the residues of  $x, y$  modulo  $k$  fixed, and dividing two such solutions in  $\mathbb{Z}[\sqrt{d}]$  gives a solution of the equation with  $k = 1$ . The fundamental solution is read off from the continued fraction of  $\sqrt{d}$ .

**Theorem 3 (infinite descent).** If a solution in positive integers produces a strictly smaller one, there is none. *Examples:*  $x^4 + y^4 = z^2$  has no solution, hence neither has  $x^4 + y^4 = z^4$ ;  $x^2 + y^2 = 3z^2$  has none, by looking modulo 3 and dividing; and  $\sqrt{2}$  is irrational.

**Theorem 4 (sums of squares).** Every positive integer is a sum of four squares (Lagrange), while the integers representable by three squares are exactly those not of the form  $4^a(8b + 7)$ , and by two squares exactly those in which every prime  $\equiv 3 \pmod{4}$  occurs to an even power (Number theory – 3).

**Fact 1 (the local-global principle and its failure).** An equation solvable in  $\mathbb{Z}$  is solvable modulo every  $n$ , which is how impossibility is usually proved. The converse holds for quadratic forms in the rational numbers (Hasse–Minkowski) but fails in the integers and for higher degrees; Selmer’s cubic  $3x^3 + 4y^3 + 5z^3 = 0$  is the standard counterexample. Finally, Thue’s theorem: an equation  $F(x, y) = m$  with  $F$  an irreducible homogeneous form of degree at least 3 has finitely many solutions.

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1. Prove Theorem 1, solve  $17x + 13y = 2026$  in integers, and prove the Chicken McNugget theorem: for coprime  $a, b$  the largest integer not representable as  $ax + by$  with  $x, y \geq 0$  is  $ab - a - b$ , and exactly  $(a - 1)(b - 1)/2$  integers are non-representable.
2. Prove that Pell’s equation is always solvable and describe all solutions; compute the fundamental solution for  $d = 13$  from the continued fraction of  $\sqrt{13}$ , and deduce that  $x^2 - 13y^2 = -1$  is solvable while  $x^2 - 34y^2 = -1$  is not.
3. Prove that there are unique positive integers  $a$  and  $n$  with  $a^{n+1} - (a + 1)^n = 2001$ . (Putnam 2001)
4. Prove by descent that  $x^4 + y^4 = z^2$  has no solution in positive integers, and deduce Fermat’s theorem for the exponent 4. Then find all Pythagorean triples and all solutions of  $x^2 + y^2 = 2z^2$ .
5. Determine all triples  $(x, y, z)$  of positive integers with  $x \leq y \leq z$  and  $x^3(y^3 + z^3) = 2012(xyz + 2)$ . (IMO shortlist 2012, N2)